

SYSTEM MODEL AND METHODOLOGY

3.1 Introduction

This chapter outlines the system architecture, dataset preparation, feature engineering procedures, and the modelling techniques used to develop an early-stage ransomware detection system based on network traffic analysis. The methodology is designed around the premise supported by existing studies that ransomware can be identified during its pre-encryption communication phase through analysis of encrypted or unencrypted flow metadata rather than payload contents

The chapter describes the workflow followed from data collection to model evaluation, including the implementation of two machine learning (ML) and three deep learning (DL) models.

3.2 System Model Overview

The proposed system is structured around a network-based detection pipeline capable of identifying ransomware indicators before encryption occurs. The detection process operates at the network perimeter where all inbound and outbound traffic passes through a monitored gateway.

3.2.1 Network Environment

The system assumes an enterprise-like architecture consisting of:

- Client endpoints, where ransomware infection may originate
- A traffic capture point positioned at a firewall, router, or gateway

- A flow generation and feature extraction module that converts packets into structured flows
- A machine learning and deep learning inference engine, which predicts whether flows are benign or ransomware-related.

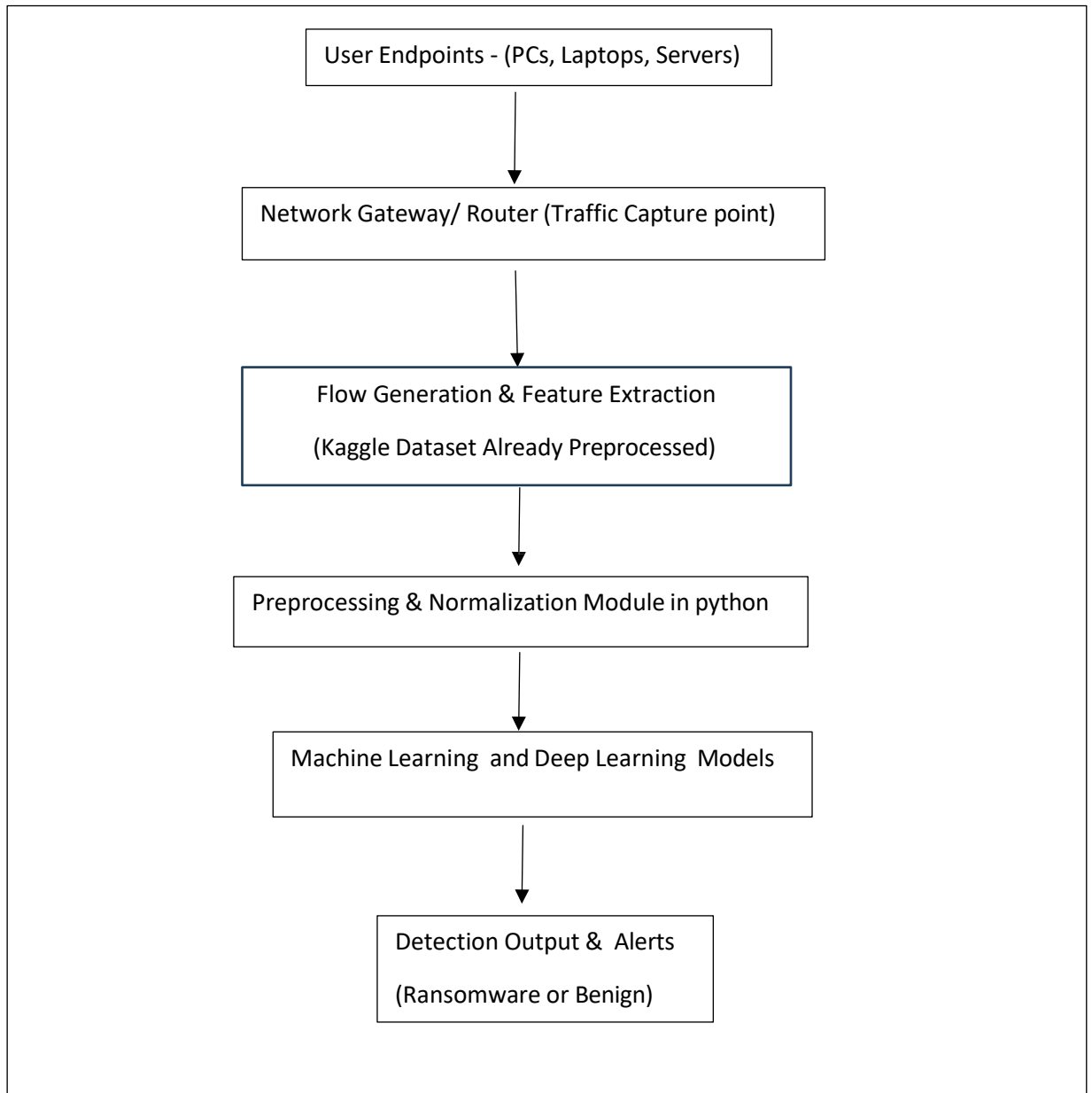


Figure 3.2.1: System architecture showing the end-to-end ransomware detection pipeline from captured network traffic to ML/DL classification.

This architecture is consistent with previous research emphasising network-layer visibility as a critical vantage point for pre-encryption detection, especially when ransomware uses encrypted C2 channels such as HTTPS/TLS .

3.3 Dataset and Flow Extraction

3.3.1 Data Source

The dataset used in this research consists of labelled benign or goodware and ransomware network flows. Each flow represents a bidirectional exchange between a host and a remote server, summarised by statistical and temporal features commonly used in encrypted malware traffic analysis.

3.3.2 Flow Generation

In this study, flow records were **not generated directly from raw PCAP files**. Instead, the dataset was obtained from **Kaggle[12]**, where the network traffic had already been pre-processed into structured flow-level records. This means that the feature extraction stage—typically performed using tools such as CICFlowMeter or Zeek—had already been completed by the dataset creators.

The Kaggle dataset consisted of numerical and categorical attributes representing statistical properties of each network flow, including:

- Flow duration
- Total forward and backward packet count
- Byte counts
- Inter-arrival time (IAT) metrics
- Packet size statistics
- Flag counts and connection metadata

These features closely resemble those used in earlier encrypted traffic detection studies, such as Barut et al. and Modi, who relied on flow-level metadata rather than raw payload contents for malware detection. As highlighted in the literature review, such features have been repeatedly validated as effective for distinguishing ransomware communications from benign traffic, even when the payload is encrypted

By using a Kaggle dataset that already includes these attributes, the study bypasses computationally expensive PCAP parsing while still leveraging the same high-value

statistical features required for ML/DL-based ransomware classification. The dataset therefore provides an appropriate and reliable foundation for training both machine learning and deep learning models.

3.4 Data Preprocessing

Data preprocessing ensures clean, consistent, no duplicate and model-ready input.

3.4.1 Cleaning and Normalisation

- Flows with missing or invalid values were removed.
- Numerical features were scaled using standard normalisation to stabilise training.
- Categorical attributes (e.g., protocol type) were one-hot encoded.

3.4.2 Label Encoding

Flows were labelled:

- 0 – benign
- 1 – ransomware (pre-encryption stage)

3.4.3 Dataset Split

A 75 – 25 stratified split was applied, ensuring proportional class representation across training and testing subsets.

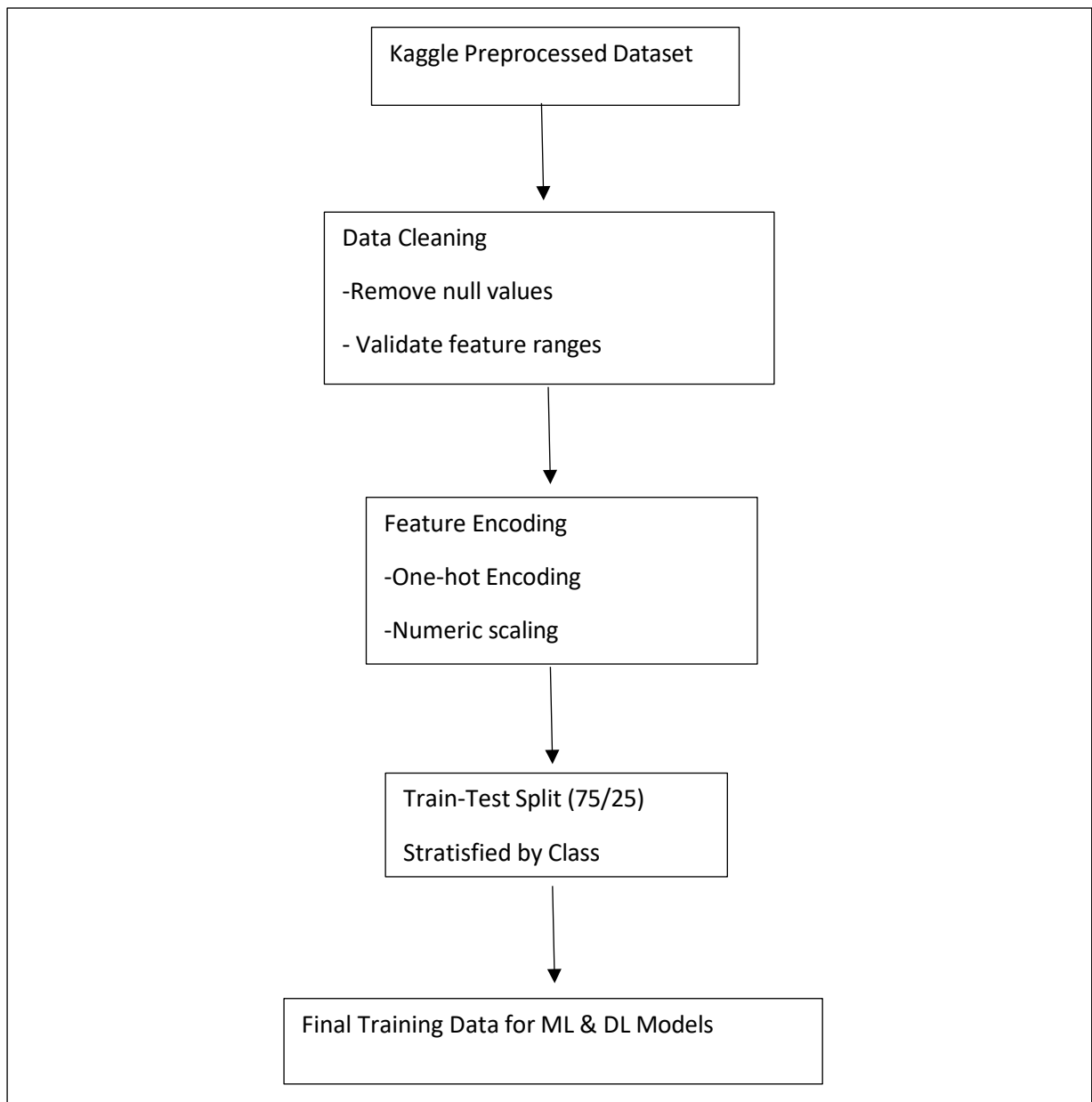


Figure 3.2: Data preprocessing workflow showing cleaning, encoding, normalisation, and preparation for training.

Five models were trained and evaluated:

Machine Learning Models

1. Logistic Regression
2. Random Forest

Deep Learning Models

1. Fully Connected Neural Network (MLP)
2. Residual Multilayer Perceptron (Residual MLP)
3. One-Dimensional Convolutional Neural Network (1D-CNN)

The combination of ML and DL models enables comparative assessment of linear, non-linear, and deep hierarchical feature extraction capabilities.

3.6 Machine Learning Models

3.6.1 Logistic Regression

Logistic Regression provides a strong linear baseline. It operates by finding optimal decision boundaries in high-dimensional feature space. Its use in malware traffic analysis is well established due to its interpretability and speed.

3.6.2 Random Forest

Random Forest is an ensemble model composed of multiple decision trees. It captures complex, non-linear patterns in network traffic and has demonstrated high performance in detecting encrypted ransomware communications in prior research (e.g., Modi's thesis).

Both ML models were implemented using Scikit-learn, with hyperparameters tuned minimally to avoid overfitting.

3.7 Deep Learning Architectures

Three deep learning models were implemented to learn abstract feature representations unavailable to traditional ML methods:

3.7.1 Fully Connected Feedforward Neural Network (MLP)

The MLP model consisted of multiple dense layers with ReLU activation and dropout for regularisation. It was trained for 30 epochs, this model displayed dramatic loss reduction from 13.56 $\rightarrow$ 0.0023 achieving stable convergence with a final AUC of 0.959 and accuracy of 97%.

The architecture follows the principle that neural networks can learn non-linear relationships between flow features that simpler models cannot capture.

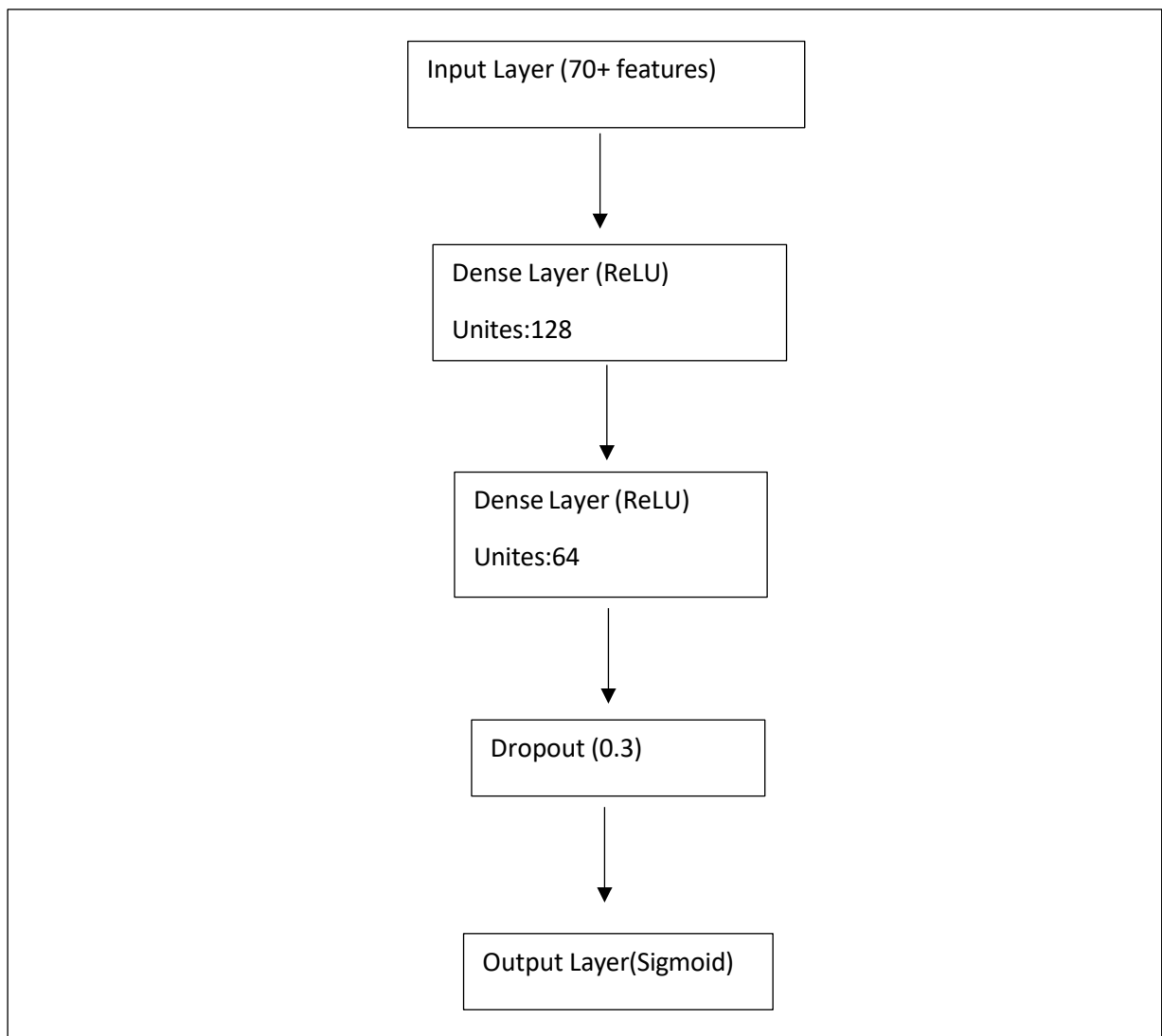


Figure 3.7.1: Architecture of the fully connected neural network (MLP) used for ransomware detection.

3.7.2 Residual Multilayer Perceptron

A Residual MLP was implemented using skip connections.

Residual learning: Improves convergence stability, Mitigates vanishing gradient issues and enables deeper architectures

This model displayed dramatic loss reduction from 10.53 $\rightarrow$ 0.0002, indicating efficient optimisation. Its AUC reached 0.9724, outperforming the standard MLP.

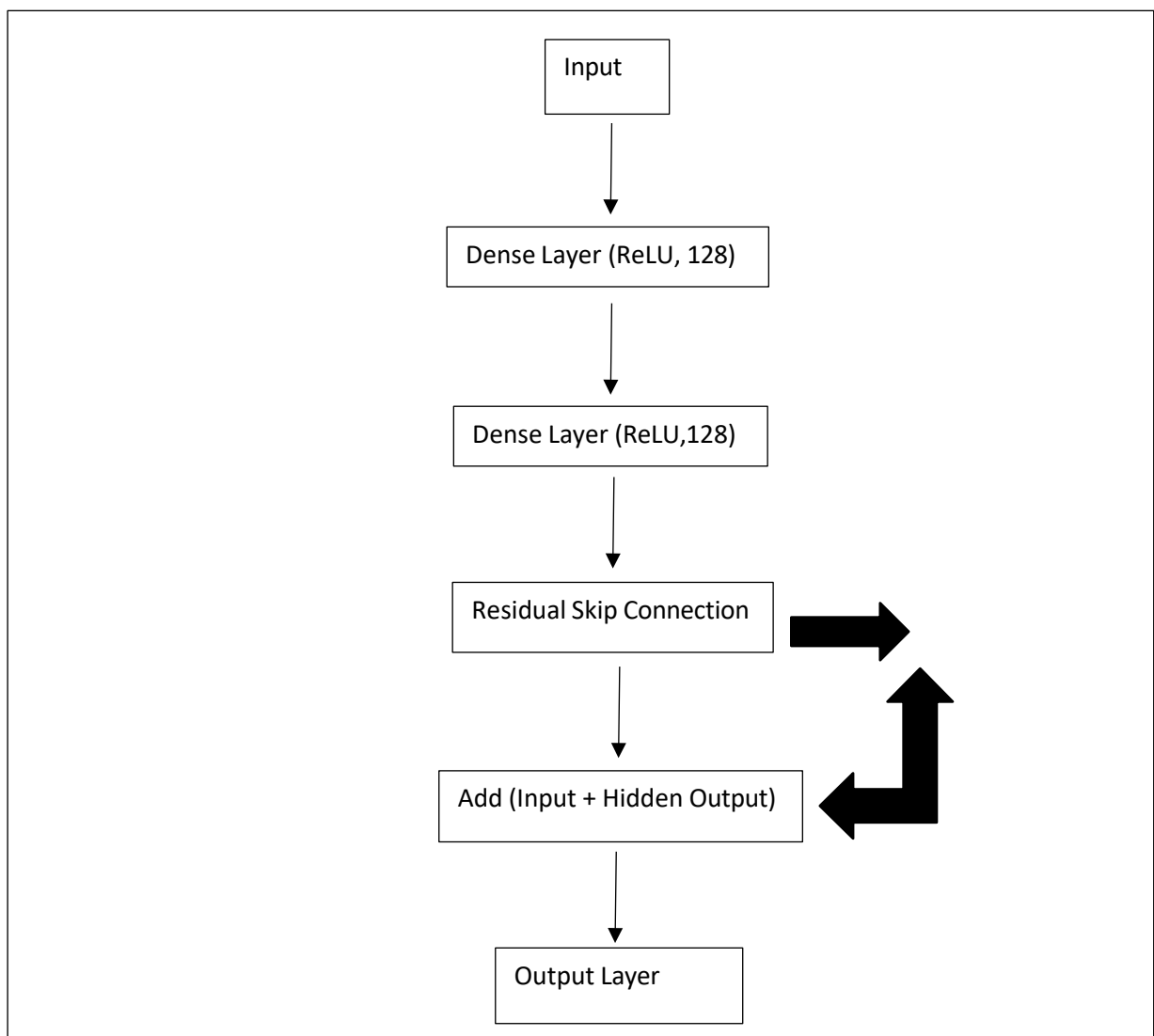


Figure 3.4: Residual MLP architecture showing skip connections that stabilise training and improve convergence.

3.7.3 One-Dimensional Convolutional Neural Network (1D-CNN)

The 1D-CNN treats flow features as a structured sequence, enabling the model to:

- Capture local dependencies
- Learn hierarchical feature abstractions
- Detect patterns analogous to C2 beaconing or staged ransomware communication

Training loss decreased from 11.65 $\rightarrow$ 0.0011, and the model achieved the highest AUC (0.9863) among all models.

This result aligns with image- based and signal-based malware detection research, where convolutional filters excel at identifying subtle structural anomalies in network traffic.

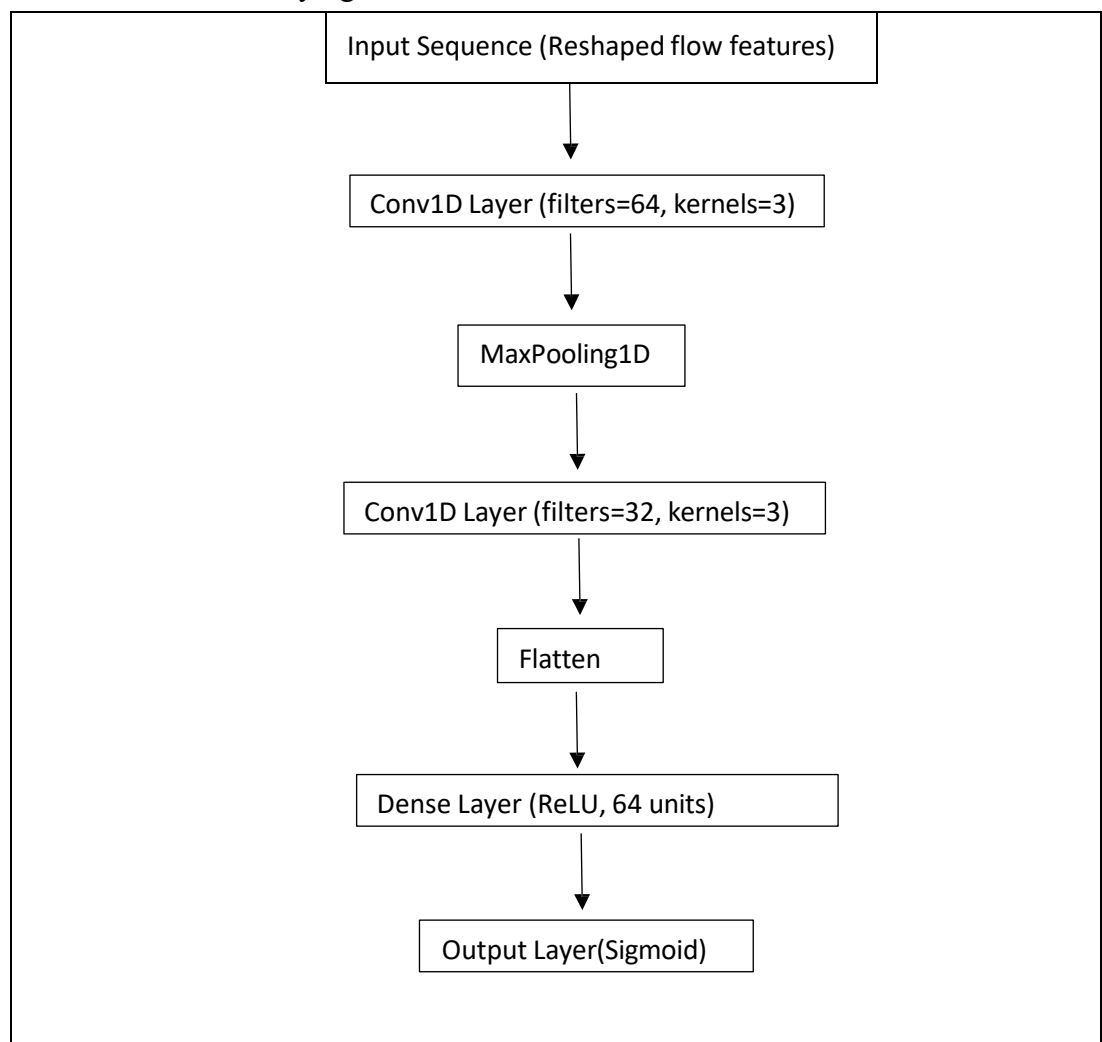


Figure 3.7.3: 1D Convolutional Neural Network architecture used to detect ransomware communication patterns.

3.8 Training Procedure

All deep learning models used:

- Binary cross-entropy loss function
- Adam optimiser
- Batch size selected to balance performance and stability
- Early stopping to minimize overfitting
- Machine learning models were trained using standard Scikit-learn pipelines.

3.9 Evaluation Metrics

Performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- AUC (Area Under ROC Curve)

The inclusion of AUC is crucial because it evaluates ranking ability across thresholds, which is particularly relevant for high-stakes security applications.

These metrics follow best practices in encrypted traffic and ransomware detection research, as recommended in prior studies.

3.10 Summary

The methodology integrates classical ML and advanced DL models into a unified detection pipeline, allowing comprehensive evaluation of their capabilities in identifying ransomware pre-encryption network behaviour. The next chapter presents the experimental results and a detailed discussion of model performance.